

EP3120 - Statistical Mechanics

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Thermodynamics

Thermodynamics

- Thermodynamics is a phenomenological theory of bulk matter
- It means that it is based directly on observations of properties of bulk matter and changes in them (“dynamics”)
- It describes the relationship in between these different observed properties, for ex., mass, volume, energy, temperature, pressure, heat capacity, etc.
- It mostly deals with properties related to heat (“thermo”), but it also encompasses other bulk properties (like polarization, conduction, magnetization, etc.)
- One of the major aims of statistical mechanics is to explain thermodynamics from fundamental physics principles

Thermodynamic limit

- The thermodynamic limit is defined as

$$N \rightarrow \infty; V \rightarrow \infty \quad (1)$$

$$\frac{N}{V} = \text{constant} \quad (2)$$

- Typically a thermodynamic system is characterized by pressure (P), volume (V), and temperature (T) - ex. an ideal gas
- These are macroscopic properties
- There may be more, or less, or different properties also, depending on the system
- For ex., for a very thin film area instead of volume, charge carrier density for semiconductors, magnetization for magnetic materials, etc.
- Extensive properties are those that are proportional to the amount of matter in the thermodynamic limit
- Intensive properties are independent of the amount of matter in the thermodynamics limit
- Question: Is ideal gas pressure an extensive or intensive property?

First law of thermodynamics

- Let ΔQ denote the net amount of heat absorbed by the system
- ΔW be the work done by this system
- There is a quantity called internal energy which obeys the relation

$$\Delta U = \Delta Q - \Delta W \quad (3)$$

- The first law states that this internal energy is a state function, it does not depend on how the transformation of the system is done - it depends only on the starting state and ending state of the system
- ΔQ and ΔW are not state functions, they depend on the path taken from initial to final state
- An equation of state relates the various macroscopic properties of the system

$$f(P, V, N, T) = 0 \quad (4)$$

- For the example of ideal gas

$$PV = Nk_B T \quad (5)$$

State variables

- Question: Is ideal gas pressure an extensive or intensive variable?
- This reduces the number of independent variables by one, i.e., P , V , and T . Often times, we can also consider N to be fixed
- This means that state variables can be expressed as a smooth differentiable function of two of these independent variables, i.e.,

$$U \equiv U(P, V) \text{ or } U(V, T) \text{ or } U(P, T) \quad (6)$$

- But you cannot write Q or W as functions of these independent variables
- So state variables are also exact differentials, i.e.

$$dU = \left. \frac{\partial U}{\partial P} \right|_V dP + \left. \frac{\partial U}{\partial V} \right|_P dV \equiv A(P, V)dP + B(P, V)dV \quad (7)$$

$$\implies \left. \frac{\partial A}{\partial V} \right|_P = \left. \frac{\partial B}{\partial P} \right|_V \quad (8)$$

Internal energy

- The internal energy represents energy from the interactions of the constituent particles in the system (their kinetic energy, rotational energy, potential energy, etc.)
- Consider a system with N fixed and the equation of state is also given. This means that the internal energy depends on two independent system variables
- If we choose them as $U(V, T)$ then an infinitesimal change in the internal energy becomes

$$dU = \left. \frac{\partial U}{\partial V} \right|_T dV + \left. \frac{\partial U}{\partial T} \right|_V dT \quad (9)$$

- From the first law of thermodynamics

$$\left. \frac{\partial U}{\partial V} \right|_T dV + \left. \frac{\partial U}{\partial T} \right|_V dT = dQ - PdV \quad (10)$$

$$\Rightarrow C_V \equiv \left. \frac{\Delta Q}{\Delta T} \right|_V = \left. \frac{\partial U}{\partial T} \right|_V \quad (11)$$

Enthalpy

- Instead, if we take $U(P, T)$ as the independent variables, then from first law of thermodynamics

$$\delta Q = \left. \frac{\partial U}{\partial T} \right|_P dT + \left. \frac{\partial U}{\partial P} \right|_T dP + PdV \quad (12)$$

- At constant pressure we can write

$$\left. \frac{\delta Q}{\delta T} \right|_P = \left. \frac{\partial U}{\partial T} \right|_P + P \left. \frac{\partial V}{\partial T} \right|_P \quad (13)$$

$$= \left. \frac{\partial}{\partial T} (U + PV) \right|_P = \left. \frac{\partial H}{\partial T} \right|_P \quad (14)$$

- Here we define the enthalpy $H \equiv U + PV$

$$C_P \equiv \left. \frac{\partial H}{\partial T} \right|_P \quad (15)$$

Second law of thermodynamics

- Q. Is Enthalpy a state variable?
- Experimentally it is found that if an ideal gas expands into vacuum ($\Delta W = 0$, $\Delta Q = 0$), then its temperature remains unchanged

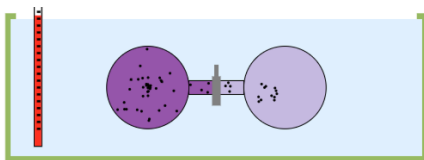
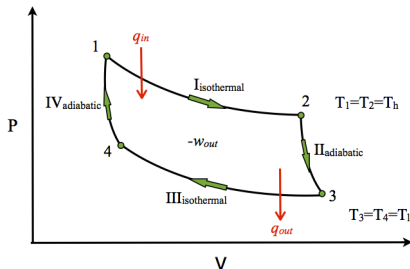


Figure: <https://demonstrations.wolfram.com/JouleExperimentOnFreeExpansion>

- This shows that the internal energy of an ideal gas depends only on temperature
- H.W.: Show that $C_P - C_V = Nk_B$ for an ideal gas with $PV = Nk_B T$
- 2nd law - There exists no thermodynamic transformation whose sole effect is to extract a quantity of heat from a cold reservoir and deliver it to a hot reservoir

Carnot engine

- An equivalent statement - There exists no thermodynamic transformation which solely extracts a quantity of heat from a reservoir and converts it entirely into work
- This is a system that exchanges energy between two reservoirs and does some work in a cycle



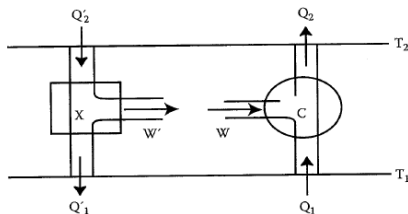
- From first law we have $\Delta W = Q_{in} - Q_{out}$
- The efficiency is defined as

$$\eta = \frac{\Delta W}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}} \quad (16)$$

Reversibility

- It can be shown that the Carnot engine is the most efficient engine
- Proof by contradiction
- Suppose another engine has more efficiency than Carnot engine, it takes heat Q'_2 , does work W' and gives out heat Q'_1
- If η_C is the efficiency of a Carnot engine, then

$$\frac{W'}{Q'_2} = 1 - \frac{Q'_1}{Q'_2} > \eta_C \quad (17)$$



Reversibility

- Now supply this work W' to a Carnot engine and run it in reverse, so that it takes Q_1 heat from T_1 and gives Q_2 heat to T_2
- A Carnot engine is reversible because both the isothermal and adiabatic processes are carried out slowly in equilibrium
- As per Carnot engines efficiency

$$\eta_C = \frac{W}{Q_2} < \frac{W'}{Q_2} \quad (18)$$

- This would imply that $Q'_2 < Q_2$
- Effectively, we have transferred $Q_2 - Q'_2$ heat from the cold reservoir to the hot reservoir without doing any work
- This violates the second law of thermodynamics - implies that Carnot engine is the most efficient engine
- This is true for any reversible engine operating between two reservoirs - conversely any irreversible engine will have lesser efficiency

Temperature

- Exercise: It can be shown that the efficiency of the Carnot engine is

$$\eta = 1 - \frac{T_1}{T_2} \quad (19)$$

- Since the efficiency cannot be larger than one, it means we cannot have negative temperatures - there is an absolute zero temperature
- The scale factor of this temperature scale is set such that the triple point of water corresponds to 273.16K - for historical purposes
- The Boltzmann constant is set accordingly

Entropy

- Clausius' theorem - In any cyclic transformation throughout which the temperature is defined, the following inequality holds

$$\oint \frac{dQ}{T} \leq 0 \quad (20)$$

and the equality holds only if the cycle is reversible

- Assume that this cycle, P , takes the system through a series of n temperatures $T_1 \rightarrow T_2 \rightarrow \dots T_n$
- Any cycle can be thought of as such by taking $n \rightarrow \infty$
- Now at each step, suppose the system takes heat Q_i at temperature T_i
- We have to show that

$$\sum_{i=1}^n \frac{Q_i}{T_i} \leq 0 \quad (21)$$

- Taking $n \rightarrow \infty$ will give Clausius' theorem

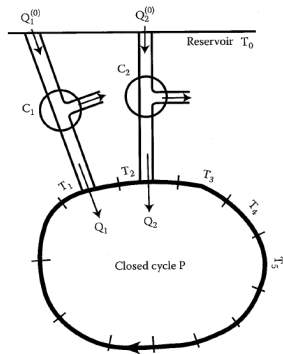
Entropy

- Now suppose at each step, a Carnot engine is also running between temperature T_0 (and $T_0 \geq T_i$ for all i)
 - It takes heat Q_i^0 from T_0 and gives heat Q_i to the system at every step T_i (it can operate in reverse also if $Q_i < 0$)
- From efficiency of Carnot engine we have

$$\frac{Q_i}{Q_i^0} = \frac{T_i}{T_0} \quad (22)$$

- In the whole cycle, this Carnot engine takes up heat

$$Q^0 = \sum_{i=1}^n Q_i^0 = T_0 \sum_{i=1}^n \frac{Q_i}{T_i} \quad (23)$$



Entropy

- During the composite cycle of P and n Carnot cycles, net heat Q^0 is absorbed from reservoir T_0
- The work done by the system is

$$W_{system} = \Delta Q - \Delta U = \sum_{i=1}^n Q_i \quad (24)$$

- The work done by the Carnot engine is

$$W_{carnot} = \sum_{i=1}^n [Q_i^0 - Q_i] \quad (25)$$

- So the net work done is

$$W_{net} = \sum_{i=1}^n Q_i^0 = Q^0 \quad (26)$$

- By second law of thermodynamics, $Q_0 \leq 0$ because otherwise we would have entirely converted heat Q^0 into work

Entropy

- Since $T_0 > 0$ we get

$$\sum_{i=1}^n \frac{Q_i}{T_i} \leq 0 \quad (27)$$

- If cycle A is reversible, then we can run this process in reverse, with the Carnot engine also running in reverse
- The heats will change signs and we will get

$$-\sum_{i=1}^n \frac{Q_i}{T_i} \leq 0 \quad (28)$$

- Since both inequalities have to be valid, hence for the reversible cycle we get the equality
- Corollary - For reversible transformations, the quantity

$$\int \frac{dQ}{T} \quad (29)$$

is a state function

Entropy

- Consider two states of the system A and B . Consider two reversible paths, 1 and 2, going from A to B
- Suppose we go from A to B along path 1, and then from B back to A along path 2 in reverse
- By Clausius' theorem we can say

$$\int_1 \frac{dQ}{T} - \int_2 \frac{dQ}{T} = 0 \quad (30)$$

- This means that for any two reversible paths, we have the equality

$$\int_1 \frac{dQ}{T} = \int_2 \frac{dQ}{T} \quad (31)$$

- This motivates the definition of a state function

$$\Delta S = \int_{\text{reversible}} \frac{dQ}{T} \quad (32)$$

Entropy

- We can also show that for any arbitrary transformation from A to B

$$\int_A^B \frac{dQ}{T} \leq S(B) - S(A) \quad (33)$$

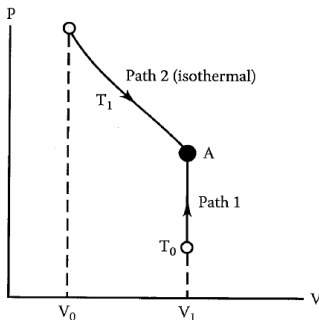
- Exercise: Show this
- Another corollary is that the entropy of a thermally isolated system never decreases
- Suppose this isolated system is in state A and goes to state B
- $dQ = 0$ because this system is thermally isolated and hence we get

$$0 \leq S(B) - S(A) \implies S(B) \geq S(A) \quad (34)$$

- So the entropy of the system cannot decrease with time

Third law of thermodynamics

- We have only defined the change in entropy as a state function - how to define the absolute value of entropy?
- Lets consider the entropy of an ideal gas at a state A in the below diagram



Absolute value of entropy

- Along path 1 we have

$$S(V_1, T_1) - S(V_1, T_0) = \int_1 \frac{dQ}{T} = C_V \int_{T_0}^{T_1} \frac{dT}{T} = C_V \ln \left(\frac{T_1}{T_0} \right) \quad (35)$$

- Along path 2 we have

$$S(V_1, T_1) - S(V_0, T_1) = \int_2 \frac{PdV}{T_1} = Nk_B \int_{V_0}^{V_1} \frac{dV}{V} = Nk_B \ln \left(\frac{V_1}{V_0} \right) \quad (36)$$

- For monoatomic ideal gas, $C_V = 3Nk_B/2$ and subtracting the two above expressions we get

$$S(V_1, T_0) - S(V_0, T_1) = Nk_B \ln \left(\frac{V_1}{V_0} \right) - \frac{3Nk_B}{2} \ln \left(\frac{T_1}{T_0} \right) \quad (37)$$

$$= Nk_B \ln \left(\frac{V_1 T_0^{3/2}}{V_0 T_1^{3/2}} \right) \quad (38)$$

Absolute value of entropy

- Re-arranging we get

$$S(V_1, T_0) - S(V_0, T_1) = Nk_B \ln(V_1 T_0^{3/2}) - Nk_B \ln(V_0 T_1^{3/2}) \quad (39)$$

- So in general we can write the entropy of a monoatomic ideal gas as

$$S(V, T) = C_0 + Nk_B \ln(VT^{3/2}) \quad (40)$$

- Question: Is this an extensive quantity?
- It does not appear so, because the second term depends on $N \ln(V)$, so when we take the thermodynamic limit keeping N/V constant, this will scale as $N \ln(N)$
- If the constant C_0 somehow has the form

$$C_0 = \gamma - Nk_B \ln(\delta N) \quad (41)$$

- Where γ and δ are constants

Third law of thermodynamics

- Then we get the entropy is

$$S(V, T) = \gamma + Nk_B \ln \left(\frac{VT^{3/2}}{N\delta} \right) \quad (42)$$

- Here we can see clearly the $\propto N$ dependence in the thermodynamic limit, making it an extensive property
- In order to make it truly extensive, the constant γ has to be zero
- However, then the expression inside the $\ln$ blows to minus infinity when $T \rightarrow 0$
- In order to fix this, the third law of thermodynamics was proposed - The entropy of a system at absolute zero is a universal constant, which may be taken to be zero
- In order to get the correct expression for the entropy at low temperatures which satisfies the third law, we will need to make use of quantum corrections to get the correct expression for entropy
- This is a striking example of quantum mechanics fundamentally affecting macroscopic properties

Thermodynamic potentials

- The Helmholtz free energy is defined as

$$A = U - TS \quad (43)$$

- The Gibbs potential is defined as

$$G = A + PV \quad (44)$$

- These potentials/free energies are useful in the study of phase transitions
- Consider a system undergo an arbitrary isothermal transition from state A to state B
- From second law we can say that

$$\int_A^B \frac{dQ}{T} \leq S(B) - S(A) \quad (45)$$

- As the process is isothermal, we have

$$\Delta Q \leq T\Delta S \quad (46)$$

Helmholtz free energy

- From the first law we can write

$$W \leq T\Delta S - \Delta U \implies W \leq -\Delta A \quad (47)$$

- For a mechanically isolated system (i.e. no work can be done on/by it) this means that $\Delta A \leq 0$ - its Helmholtz free energy cannot increase
- The equality holds for reversible process, in that case its Helmholtz free energy remains unchanged
- For a mechanically isolated system kept at constant temperature since $\Delta A \leq 0$ it means that if the system is in a (local) minimum Helmholtz energy state, then it is in equilibrium
- Now consider a system kept at constant pressure and temperature
- Since T is kept constant, so

$$\Delta Q \leq T\Delta S \implies W \leq -\Delta A \quad (48)$$

Gibbs thermodynamic potential

- Since pressure is constant, so $W = P\Delta V$
- Therefore we get

$$P\Delta V + \Delta A \leq 0 \implies \Delta G \leq 0 \quad (49)$$

- As before this means that a system is at a (local) minimum in Gibbs potential, at constant P and T , then it is in equilibrium
- These potentials are useful in analyzing systems made up of more than one component, which need to come into equilibrium

Maxwell relations

- Exercise: Show that in a reversible transformation

$$dG = -SdT + VdP \quad (50)$$

- From this we can write

$$S = -\left.\frac{\partial G}{\partial T}\right|_P; \quad V = \left.\frac{\partial G}{\partial P}\right|_T \quad (51)$$

- Similarly, it can be shown that

$$dH = TdS + VdP \quad (52)$$

- From this we get

$$T = \left.\frac{\partial H}{\partial S}\right|_P; \quad V = \left.\frac{\partial H}{\partial P}\right|_S \quad (53)$$

Maxwell relations

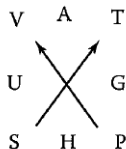


Figure 3.6 Mnemonic diagram summarizing the Maxwell relations. Each quantity at the center of a row or column is flanked by its natural variables. The partial derivative with respect to one variable, with the other kept fixed, is arrived at by following the diagonal line originating from that variable. Attach a minus sign if you go against the arrow.